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The Editors in Chief

Electrical Engineering

Springer

Dear Editors in Chief,

I am pleased to submit the manuscript “When does decision-focused learning add value to residential battery control? Private costs and AC feeder consequences” for consideration as an original research article in Electrical Engineering. It addresses a concrete problem in residential battery control: the household controller already optimizes an announced tariff, so the question is whether training its signal against realized operating cost, rather than forecast error, repays the training and maintenance it requires, and what the changed battery action does to voltage on the feeder that hosts it.

Earlier studies establish that forecast error and operating value are different quantities, and they supply the differentiable optimization methods that make cost-based training practical. Residential storage studies show that these models compete with conventional alternatives rather than dominating them. The incremental engineering value has not been established. It is unclear how much of the gain survives against a low-dimensional calibration, against explicit scenario uncertainty, or against a changed deployment input, nor what that gain costs the network. Feeder studies usually optimize a system service, whereas the learner examined here receives no voltages and no network constraints, so any voltage change is a by-product of a private objective.

To answer these questions I hold initialization, data, update budget and checkpoint selection fixed and compare a decision loss with continued squared-error training, using measured household consumption and PV traces, simulated batteries and public AC feeder models that include an unbalanced European low-voltage benchmark and two SimBench networks. Deployable alternatives are compared on the same information: three trainable cost offsets, direct net-load prediction and a residual scenario controller with 20 scenarios. A frozen transfer to ten independent Irish PV households tests how far the signal travels, and sensitivities cover source voltage, wear price, continuous state of charge and an export limit.

Across 108,641 paired household-days, the decision loss lowers operating cost by 1.131 Australian cents per day relative to continued squared-error training, and three trainable offsets already recover 54.1% of that gain. A controller using 20 scenarios reaches a further 2.462 cents below the decision loss. Voltage consequences depend on the feeder: matched screening-band exposure falls by 0.853 percentage points at a 1.050 pu source and by 0.00848 points at 1.025 pu, while the two alternative networks show no screening-band events at all. Frozen transfer to the ten Irish households gives a 1.252-cent per paired-day time-of-use reduction. Matched persistence retraining saves about 1.6 cents per owner-day while overvoltage exposure rises by 0.373 and 0.606 percentage points, so a private saving can arrive with a measurable network penalty.

These results speak directly to work on residential battery control, low-voltage distribution networks, voltage screening and method selection. The paper identifies when a three-parameter calibration carries most of the benefit, when scenario control is the better engineering choice, and when household savings accompany a worsening of feeder voltage, and it traces the effect from the altered battery action to the AC voltage a feeder experiences. Reviewers with expertise in distribution-network

voltage analysis, residential storage scheduling, probabilistic forecasting or decision-focused learning would be well placed to assess the work.

The manuscript is original, is not under consideration elsewhere, and as its sole author I have approved this submission. The paper lets an engineer choose, before committing to a task-trained controller, between low-dimensional calibration, scenario control and a learner trained on operating cost for a given tariff, dataset and feeder, and anticipate the voltage consequence of that choice.

Sincerely,

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